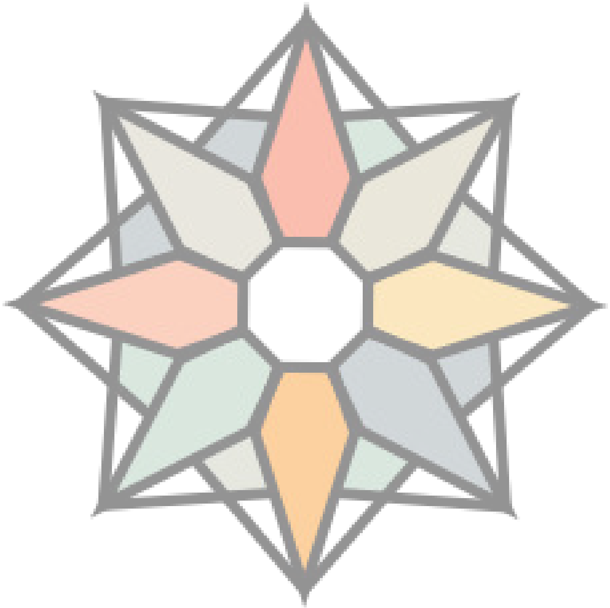


Stained Glass Panels

Hackathon Grasshopper Users Meeting

Ewa Romanczuk

Bartlomiej Tokarzewski

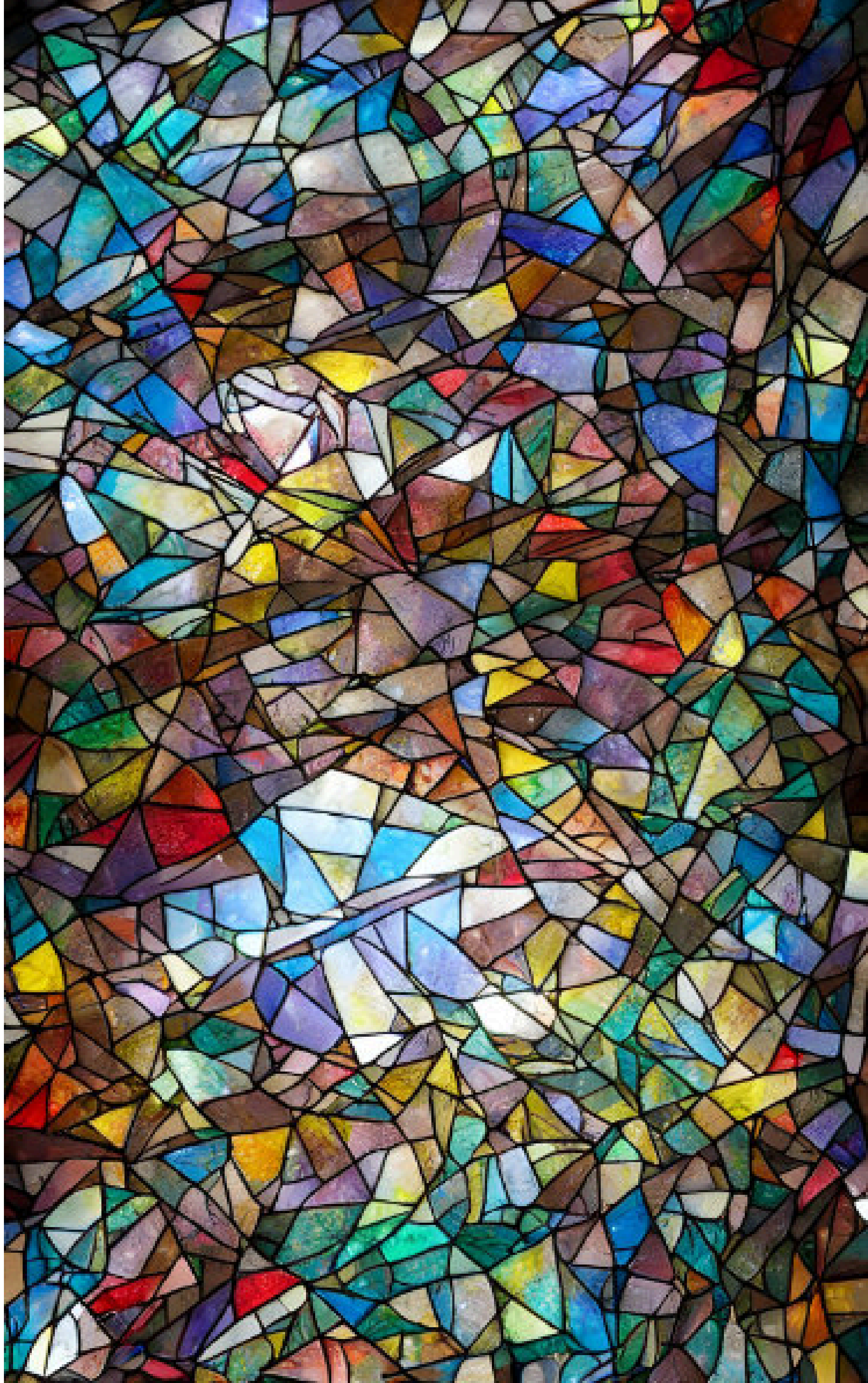
Martyna Hypta

Mateusz Tokarczyk

Somia Hassan

Content

- The Problem
- The Vision
- The Inputs
- The Workflow
- The Tools
- Live Demo
- What's Next



The Problem

We want unique and beautiful elements in our spaces.

Problem: Custom ornament is expensive because it requires intense manual design and specialized craftsmanship.

Result: Budget constraints force us back into sterile, mass-produced minimalism.

Our Solution

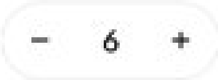
We move the complexity from manual labor to software.

Platform: A configurator that lets anyone turn an image into a stained glass panel.

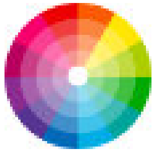
Output: Instant, 3D-printable files. No specialized craft skills required; no fortune needed.

The Inputs

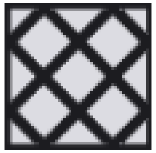
Raster



Number of colours



Colours



Pattern of the panels

The Outputs

Visualisation

Model 3D in .glb

The workflow

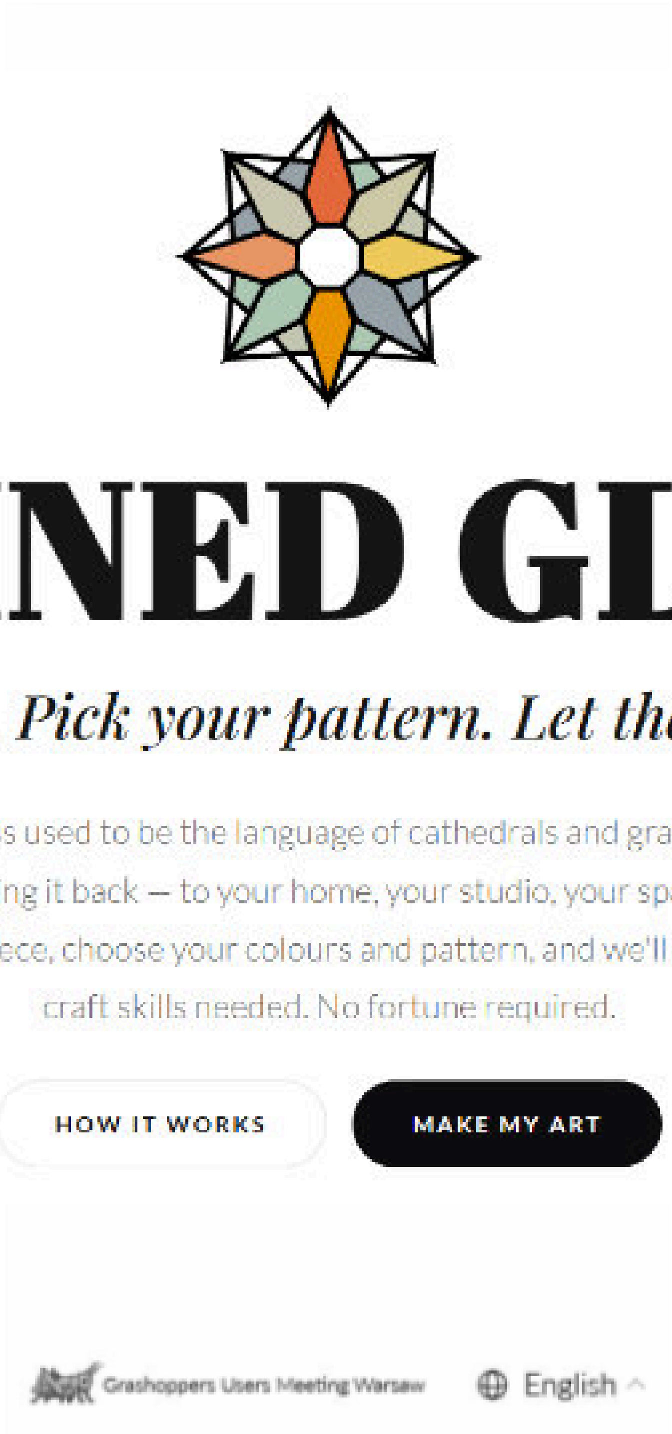
Parallel tracks from the start. We split the team into:

- image-processing - Somia, Mateusz
- geometry - Martyna, Ewa
- web/render/deploy - Bartłomiej
- UI and UX design - Martyna

The Tools

Rhino 7 + Grasshopper	First geometry engine — image sampling, colour quantization, panelization, BREP frames/extrusion
Python / Jupyter	Early model transport from Grasshopper to the web viewer
Rust + wasm-bindgen	The production engine, compiled to WebAssembly (sg-wasm/)
three.js (WebGPU)	The 3D "beauty" renderer
Cloudflare + Wrangler	Hosting (no-build static site)
Illustrator/Photoshop	Logo (star rosette), grasshopper mascot, loading GIF, app mockups

Live Demo


STAINED GLASS
Pick your colours. Pick your pattern. Let the light do the rest.
Stained glass used to be the language of cathedrals and grand estates. We're bringing it back — to your home, your studio, your space. Design your own piece, choose your colours and pattern, and we'll print it. No craft skills needed. No fortune required.
HOW IT WORKS
MAKE MY ART
Grashoppers Users Meeting Warsaw
English

What's Next?

Our prototype can develop in multiple ways, we propose this roadmap to advance our Idea:

1) Create viable way of connecting panels to each other using Printable frame exported using website

2) Line recognition of pictures uploaded on website

3) Optimised technical view with detail of connections and 2d drawings

4) The printer selection

5) Estimated price of the finished product

6) Gallery of the pictures with finished model

7) Online shop for special orders and infrastructure

8) 3D printer farms for orders

9) Metal printing of the mesh

Thank you!